

HAGERSTOWN, MD, USA

WMO: 724066

Lat: 39.706N

Lon: 77.730W

Elev: 215

StdP: 98.77

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 93706

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-9.1	-20.2	0.6	-9.7	-17.8	0.8	-8.0	12.7	-0.7	11.7	-1.3	4.2	290	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	33.4	22.7	32.0	22.4	30.7	21.8	24.6	30.4	23.8	29.3	23.1	28.1	3.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.7	17.9	26.9	22.2	17.4	26.5	21.5	16.6	25.9	75.2	30.7	72.1	29.4	69.4	28.3	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.2	8.0	-14.1	35.3	2.1	1.8	-15.6	36.6	-16.8	37.6	-18.0	38.6	-19.5	39.9		
(5)			-15.1	26.1	2.0	1.5	-16.6	27.1	-17.8	28.0	-18.9	28.9	-20.4	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	-0.1	1.3	6.0	12.5	17.9	22.5	24.8	23.8	20.1	13.6	7.6	2.4		
	DBStd	9.83	5.70	5.31	5.51	5.06	4.43	3.35	2.79	2.81	3.79	4.60	4.61	4.97		
	HDD10.0	1102	316	247	150	29	1	0	0	0	17	99	242			
	HDD18.3	2690	571	476	383	187	65	6	0	1	24	161	322	494		
	CDD10.0	2107	3	4	26	103	246	376	457	428	302	127	28	7		
	CDD18.3	655	0	0	1	11	52	132	199	171	77	13	1	0		
	CDH23.3	5570	0	2	7	118	423	1112	1836	1425	557	88	2	0		
	CDH26.7	1899	0	0	0	31	121	363	708	492	168	17	0	0		
(14) Wind	WSAvg	3.2	3.8	3.8	4.0	3.7	3.0	2.9	2.5	2.4	2.6	3.1	3.3	3.4		
(15) Precipitation	PrecAvg	1012	66	55	89	91	96	89	92	88	99	82	71	86		
	PrecMax	1493	97	138	186	204	196	220	156	165	209	209	112	150		
	PrecMin	663	31	7	10	45	30	29	44	40	0	12	32	36		
	PrecStd	184	20	29	41	42	44	50	33	34	56	53	25	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	20.4	24.2	30.2	32.0	33.8	35.9	34.4	33.3	28.9	22.3	19.0		
		MCWB	12.7	14.0	15.3	18.0	21.5	22.7	23.5	22.7	22.2	21.0	15.5	14.8		
	2%	DB	14.0	15.8	20.7	26.7	29.8	32.1	33.7	32.7	30.8	26.0	19.0	15.7		
		MCWB	10.7	10.5	13.5	16.6	20.2	22.2	23.4	22.7	21.4	19.0	13.5	12.5		
	5%	DB	10.7	12.1	17.6	23.7	27.8	30.3	32.0	31.1	28.4	23.3	17.3	12.2		
		MCWB	7.6	7.5	11.6	15.1	19.4	21.8	22.6	22.3	20.3	17.5	12.9	9.1		
	10%	DB	7.6	8.8	15.1	21.1	25.8	28.7	30.5	29.4	26.6	21.2	15.1	9.4		
		MCWB	4.8	5.1	10.0	13.7	18.6	20.9	22.1	21.4	19.3	16.2	10.8	6.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	15.3	16.8	19.9	23.0	24.5	26.1	25.5	24.1	22.4	17.3	15.6		
		MCDB	16.3	19.2	21.2	26.5	29.1	30.6	31.3	31.2	30.2	26.1	19.5	18.2		
	2%	WB	11.4	10.6	14.8	17.9	21.7	23.5	24.8	24.3	22.9	20.6	15.3	12.7		
		MCDB	13.5	14.2	19.1	24.3	27.8	29.4	30.8	29.8	27.5	24.9	18.3	14.9		
	5%	WB	7.7	8.0	12.7	16.5	20.6	22.8	24.0	23.5	22.0	18.9	13.6	9.4		
		MCDB	10.2	11.3	17.2	21.9	25.8	28.6	29.7	28.5	25.8	22.4	16.4	11.8		
	10%	WB	5.1	5.5	10.1	14.9	19.7	22.0	23.3	22.7	21.0	16.8	11.5	6.7		
		MCDB	7.6	8.6	14.5	20.1	24.5	27.1	28.3	27.4	24.7	20.1	14.4	9.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.3	9.7	10.8	10.3	9.9	9.9	9.7	9.9	9.5	8.7	7.3		
		MCDBR	11.3	12.8	13.4	14.3	12.5	11.4	11.4	11.2	12.1	12.3	11.7	10.6		
	5% WB	MCWBR	8.7	9.1	8.5	7.3	5.7	4.7	4.0	4.1	4.8	6.3	7.8	8.6		
		MCWBR	10.3	11.7	12.5	12.3	11.1	10.0	9.7	9.5	9.1	10.0	10.1	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.324	0.355	0.390	0.430	0.493	0.515	0.510	0.430	0.366	0.322	0.308		
	taud		2.438	2.363	2.364	2.295	2.207	2.065	2.019	2.013	2.214	2.398	2.489	2.466		
	Ebn at Noon		861	892	899	885	852	797	774	764	808	833	838	831		
	Edh at Noon		81	100	111	128	142	164	171	167	129	96	77	73		
(44) All-Sky Solar Radiation	RadAvg		1.86	2.75	3.67	4.73	5.29	5.86	5.81	5.17	4.22	3.06	2.17	1.61		
	RadStd		0.16	0.33	0.23	0.40	0.56	0.46	0.34	0.27	0.55	0.35	0.20	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-215	+228	+166
(47)	Regional (60 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+62

Nomenclature: See separate page